NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Assessment Spring 2026

Level: Bachelor | Year : 2026

Program: BECE (M_D) | Full Marks: 100

Course: : **Computer Networks** | Pass Marks: 45

Semester: VI | Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a. List out the roles of ISP's and NSP's. Draw the diagram of peer to peer and client-server architecture model and differentiate. [8]

   b. Explain each layer of TCP/IP in brief and write down the importance of layer wise architecture in internet world. [7]

2. a. "Optical fibers are best and most popular among the transmission media" Justify. How is circuit switching different from packet switching? [8]

   b. How are size of windows and delays interrelated during data communication? Explain the working principle of GO BACK N ARQ process of error handling. [7]

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3. a. Illustrate with the help of an example the Cyclic Redundancy Check method at both receivers and transmitters end. [8]

   b. Draw the flowchart of CSMA/CD. How is CSMA/CA different from CSMA/CD. [7]

4. a. Given an IP address of 192.168.10. 0 / 24 create subnetworks for four departments with required hosts 106, 6, 61 and 9 respectively. Find out the subnetwork address, broadcast address, and ranges of host addresses, subnetwork masks and wildcard masks for each department. [8]

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b. Differentiate routing algorithms from different perspective. Explain with an example the distance vector routing protocol. [7]

5. a. Compare IPv6 and IPv4 Header. Mention the types of transition strategies applied for IP4v to IPv6 based packet forwarding. [8]

b. Compare Transmission Control Protocol (TCP) header and User Datagram Protocol (UDP) header format with neat diagram. [7]

6. a. What is congestion? How TCP manages congestion? Explain the concept of token based leaky bucket algorithm for congestion control. [8]

OR

Explain the process of NAT (Network Address Translation), ARP (Address Resolution Protocol) and RARP (Reverse Address Resolution Protocol).

b. What is Network Management System? Write down the steps to create a public and private key pair with the help of RSA algorithm. [7]

OR

Explain the process of cryptography and explain the key exchange procedure with an example in symmetric cryptography. List some of the malicious things intruders can do.

7. Write short Notes on (Any Two) [5 X 2 = 10]

a) ISDN
b) VPN and Firewall
c) DNS.

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Lumbini Engineering College
(Final Internal Exam)
Semester – fall

Level: Bachelor
Programme: BE Computer 6
Course: Computer Networks

Year : 2083
Full Marks : 100
Pass Marks : 45
Time : 3 hrs

Candidates are required to give their answer in their own words as far as practicable.
The figure in the margin indicates full marks.
Attempt all the questions.

1. a) Explain different topological models in Computer network. 8
   b) Discuss the TCP/IP protocol stack with a suitable example. Also compare TCP/IP with OSI reference model. 7

2. a) Explain the term bandwidth, throughput and latency. Explain the working principle of satellite communication. 7
   b) Compare Circuit switching and packet switching. Draw necessary diagram to illustrate. 8

3. a) Explain channel access mechanism in CSMA/CD. 8
   b) What is error control? Show how does forward error control(FEC) technique helps to detect and correct the error. 7

4. a) Explain HTTP protocol. Describe the difference between SMTP, POP and IMAP servers. 7
   b) The existing network of Lumbini province (172.31.255.0/22) is to be divided into network of 4 different regional network. Among which two regional network need to be subdivided into 4 subdivisional network. Provide a complete IP address plan which includes network address, broadcast address, Usable IP Pool, Subnet mask and wildcard mask. 8

5. a) Differentiate between distance vector and Link state routing. 7
   b) What is CRC? Explain in brief about sliding window protocol. 8

6. a) What is choking? Classify routing algorithms and explain in brief about each of them. 8
   b) What is traffic shaping? Briefly explain about Leaky bucket and Token bucket Algorithm. 7

7. Write Short notes on any two (2X5)
   a. VPN b. TCP Socket c. SMTP and POP

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2083/03/2			
Level	BE	Full Marks	50
Programme	BEIT, BCE	Time	
Semester	VI	1.5 hrs	

Subject: - Computer Network

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions.*
- ✓ *The figures in the margin indicate Full Marks.*
- ✓ *Assume suitable data if necessary.*

1. What is computer networking? What are the components of computer networking? Explain the application of computer networking. [7]
2. Explain OSI reference model in detail. [8]
3. What are the functions of physical layer? What is circuit switching? How is circuit switching different from packet switching? Give examples of the cases which use circuit switching and packet switching. [7]
4. What is flow control in datalink layer? Explain any two flow control techniques. [8]

   OR

   Explain pure and slotted ALOHA in medium access control in datalink layer.
5. What is error control in datalink layer? Explain CRC and Parity Methods for error detection. [7]
6. Imagine yourself as a private consultant hired by a company to set up the network for their enterprise. You are given the IP range as 192.168.10.25/24. There are five departments with hosts 54,62,32,25 and 12. There are 3 (router to router) point-to-point links also in the setup. Allocate /30 networks to such point-to-point links. Allocate the proper IP address subnets to minimize the IP address wastage. [8]
7. Write short notes on: (*any two*) [2*2.5=5]
   a. Bandwidth delay product
   b. Broadcast and Collision domains
   c. VLANs
   d. MAC addressing

B-5003

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2083/04/11			
Level	BE	Full Marks	50
Programme	BEIT, BCE	Time	
Semester	VI	1.5 hrs	

Subject: - Computer Network

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions.*
- ✓ *The figures in the margin indicate Full Marks.*
- ✓ *Assume suitable data if necessary.*

1. Explain link state routing protocol in detail. [8]

   OR

   Explain the IPv4 header structure.

2. What is TCP connection? Explain TCP connection establishment and connection termination with handshakes diagrams. [7]

   OR

   Explain TCP header structure.

3. Explain TCP congestion control mechanism. [8]

4. Explain SNMP protocol with the concepts of agent and manager. [7]

5. What is cryptography? Explain public and private key cryptography. [8]

6. What is DNS resolution? Briefly describe the different resolution methods. [7]

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   OR

   What is HTTP? Explain HTTP protocol.

7. Write short notes on: [2*2.5=5]
   a. VPN
   b. IMAP vs POP
   c. DHCP

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NEPAL ENGINEERING COLLEGE

Level: Bachelor | Semester: Spring | Year: 2026
Program: BE | Full Marks: 100
Course: Computer Network_ New | Pass Marks: 45
Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1	a)	Define computer network. Explain advantages/disadvantages. Compare Client-Server and Peer-to-Peer.	7
	b)	Explain OSI model and functions of each layer. Compare TCP/IP with OSI.	8
2	a)	Explain the concept switching & its types.	7
	b)	Explain ISDN signaling and its architecture.	8
3	a)	Describe the LLC and MAC sublayers of the Data Link layer. Explain framing and flow control mechanisms.	7
	b)	Discuss CSMA/CD and its role in Ethernet networks.	8
4	a)	Explain IPv4 addressing, subnetting, and CIDR. Compare IPv4 and IPv6 headers.	7
		OR	
		Create 16 Subnet of given IP address 192.168.10.0/24 with new subnet mask and find out the network ID, Wildcard Mask, Usable Host IP & Broadcast Mask.	
	b)	Describe distance vector and link-state routing algorithms. Provide examples of RIP and OSPF.	8
5	a)	Explain the services and features of TCP and UDP. Discuss the concept of socket programming.	7
	b)	Discuss congestion control mechanisms and traffic shaping algorithms (Leaky Bucket, Token Bucket).	8
		OR	
		Explain 3-way Handshaking in Transport layers	
6	a)	Define symmetric key and asymmetric key cryptography. Write an Algorithm of RSA	7
	b)	Describe the role of firewalls & its types and digital certificates in network security.	8
7		Write Short notes (Any Two) 2*5	
	a)	Gateway	
	b)	E-Mail Services	
	c)	ATM	

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Madan Bhandari College of Engineering

Programme: BE
Course: Computer Networks

Full Marks: 60
Pass Marks: 30
Time: 2 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Discuss the merits and demerits of computer network. Compare star, bus, ring and mesh topology.
   b) Explain the working principle of different types of network devices: hub, switch, bridge, and router.

2. a) Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter.
   b) What is error control? Explain CRC method of error detection with a suitable example.

3. a) Explain Gigabit Ethernet with the help of the following parameters: MAC sub-layer, full duplex mode, half duplex mode, topology, implementation and encoding. (7)
   b) What are the drawbacks in IPv4? Which of the drawbacks does IPv6 solve? Explain.

4. a) Perform the subnetting of 192.168.0.0 using CIDR value /26. What is VLSM and why do we use it?
   b) Discuss Interior and Exterior Routing Protocols. What is the count-to-infinity problem in Distance Vector Routing?

5. a) What is congestion in a network? Explain the congestion control mechanism.
   b) List out and explain TCP client/server socket flow with a suitable diagram.

6. a) Explain the roles of proxy server. Compare HTTP with HTTPS.
   b) What do you mean by network security? Explain the RSA algorithm.

7. Write short notes on (Any Two)
   a) Kerberos
   b) Optical Fiber
   c) CSMA/CD

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Universal Engineering and Science College
Chakupat, Lalitpur
Pre-Board Examination

Level: Bachelor
Program: BE
Course: Computer Network

Semester: Spring

Year: 2026
Full Marks: 100
Pass Marks: 45
Time: 3 hrs

Candidates are required to answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain the merit and demerit of computer networks, list the merits of client-server model over peer-to-peer model. 7

   b) Define protocol. Why do we need layered protocol architecture? Discuss each layer of TCP/IP protocol architecture in detail. 8

2. a) Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter. 7

   b) If the 7- bit Hamming codeword received by a receiver is 1011011. Assuming the even parity, state whether the received codeword is correct or wrong. If wrong, locate the bit in error. 8

   OR

   Generator is x^4+x+1 and data 1101011011 find CRC value and check for error.

3. a) Compare CSMA/CD and CSMA/CA with the help of flow chart. 7

   b) Imagine yourself as a skillful computer network administrator. Suppose there are four departments namely A,B,C,D that required 120, 30, 45, 10 hosts respectively. Given a network 192.168.2.0/24, perform subnetting in such a way that IP wastage in each network is minimum. Find out the subnetwork address, broadcast address, and range of host addresses, subnetwork masks and wildcard masks for each department. 8

4. a) What is classful and classless addressing? Differentiate between link state and distance vector routing protocol. 7

   b) What is TCP? Explain TCP connection and termination with suitable diagram along with its TCP state while connection and termination. 8

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5. a) What is congestion in a network? Explain about congestion control mechanism. 7

   b) What is the role of URLs and Web Browser in WWW. Explain about DHCP server in details. 8

6. a) What do you mean by Asymmetric key algorithm? Explain RSA algorithm Encryption and decryption process with suitable example. 8

   b) What is NMS? Explain about major components of Network Management Architecture. 7

7. Write short notes on **(Any two)** 2×5
   a) IPv4 vs IPv6
   b) SMTP
   c) Optical Fiber

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National Academy of Science and Technology

Affiliated to Pokhara University

Pre University Examination

Level: Bachelor | Semester – Spring_VI | Year : 2026
Programme: BE Computer | Full Marks: 100
Course: Computer Network | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How has computer networks changed the lives of people in networking and communication system? Illustrate with proper examples. 7

   b) Describe intra–net, extra-net and inter-net along with their usage, scope and need. 8

2. a) What are the factors that network designer must consider to establish the topologies of a Network? State and explain client server network model. 8

   b) What is flow control? Describe Stop-and-Wait, and Selective Repeat protocols with suitable diagrams. 7

3. a) What are intermediate devices in a network? Specify some of the physical, datalink and network layer devices. Describe how fibre optics cable provides faster, reliable and efficient means for data transmission 7

   b) Explain the TCP/IP model with the protocols and functions of each layer. 8

4. a) Classify HDLC- stations and state their characteristics. Explain Flag Bytes with byte stuffing framing method. 7

   b) Describe the IPv4 Frame format. 8

5. a) An organization is planning to design 4 different department. Each department consists of 50 host id with IP address192.108.111.0/27. Explain the process how will you allocate the IP address and subnet for each of the departments using above IP. 8

OR

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Decode and define IP-address based internet N/W information center. How unicast, broadcast and multicast addresses identify a host.

b) Define congestion control. Explain Leaky bucket and Token bucket algorithm in detail with a suitable example. 7

6. a) Describe the technical aspect of assigning the IP address using DHCP. Enlist some of the client and server tools used in a network. 8

OR 7

What are the function of transport layer. Explain the concept of socket programming in the transport layer protocol.

b) What is data encryption? Discuss the role of symmetric and asymmetric key cryptography with a suitable example.

7. Write short notes on any two: 2×5

a) Next Generation Network

b) 4G vs 5G

c) CSMA/CD

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Pokhara University

Everest Engineering College

Final Internal Assessment
Spring-2026

Level: Bachelor — **F.M.** 100
Program: BE CMP(6th Sem) — **P.M.** 45
Section: A/B — **Time:** 3hrs
Subject: Computer Network

Attempt all the questions.

1 a) Explain OSI reference model with its layer and its functions. Compare it with TCP/IP model. 8

b) Define Computer Network and discuss their merits and demerits. Compare PAN, LAN, MAN and WAN. 7

2 a) Discuss guided and unguided transmission media with examples. Explain the concept of Circuit Switching and Packet Switching. 7

b) What are the responsibilities of Physical Layer. Explain the services of ISDN signaling and architecture. 8

3 a) Describe the LLC and MAC sublayers of the Data Link layer. Explain framing and flow control mechanisms. 8

b) Explain an algorithm for CSMA/CD with the help of Flowchart. 7

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4 a) The APNIC Pool for Pokhara University (103.16.32.0/22) is to be divided into network of 7 different schools. Among seven schools three school need to be subdivided into two different departments. Provide a complete IP address plan which include network address, broadcast address, usable IP pool, subnet masks and wildcard masks. 8

b) Draw header format of IPV4 and mention 3 strategies of transition from IPv4 to IPV6 7

OR

Describe Routing and its types. List out any 5 differences between Distance Vector Routing and Link State Routing algorithm

5 a) Explain the services and features of TCP and UDP. Discuss the concept of socket programming. 7

b) Discuss working principles of traffic shaping algorithms (Leaky Bucket, Token Bucket) 8

OR

Explain in details the TCP connection Establishment with proper diagrams.

6 a) Discuss symmetric key (DES) and asymmetric key (RSA) cryptography. 8

b) What is Network management System? Write down the steps to create a private Public Key pair with the help of RSA algorithm along with examples. 7

7 Write short notes on: (Any two) 2*5 =10

a) VPN and Firewall
b) NGN
c) POP and IMAP

Good Luck

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POKHARA ENGINEERING COLLEGE

Level: Bachelor | Semester – Spring | Year : 2026
Programme: BE | Full Marks: 100
Pass Marks: 45
Course: Computer Networks | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

Assessment Examination.

1. a) Define Computer Network. Differentiate between Client Server and Peer to Peer Model. 1+6
   b) Define protocols, interfaces and services. Discuss in brief about the layers of OSI model. 3+5

2. a) Differentiate between guided and unguided medium. Explain the working of optical fibre. 2+5
   b) What do you understand by flow control? Explain about the operation of Go Back N ARQ and Selective Repeat ARQ method. 2+6

3. a) Compute the CRC for bit sequence 110101 using the generator polynomial x^3+x+1. Show how the error is detected when the second bit is flipped. 8
   b) Why is the world moving towards IPV6 addressing? Explain the working of Distance vector routing protocol. 2+5

4. a) Design an IP addressing plan for a university with four departments needing 200, 100, 50 and 20 hosts using VLSM. 8
   b) What is a Socket? Explain about TCP socket call in brief.

5. a) How can we control congestion in the network? Explain about Token Bucket algorithm in detail. 2+6
   b) Which ports do HTTP and HTTPs use to browse the internet? Explain the working of Telnet. 2+5

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a) In an RSA cryptosystem, p = 3, q = 11 and public key e = 7. Compute the private key d. 8

b) Define VPN. Explain different firewalls used in Computer Network. 1+6

Write short notes on **(Any Two)**: 5*2

a) Diffie Hellman
b) Router and Switch
c) Hamming Code

**********Best of Luck**********

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Computer Network, 24 July - Spring 2026

Total marks: 100, Pass marks: 45

BoCE VI semester

Attempt all the questions.

1. a) Define computer networking. Explain the advantages and disadvantages of computer networks. [2+5]

   b) Compare OSI and TCP/IP models. Explain the functions of each layer of the TCP/IP model. [3+5]

2. a) What is transmission media? Explain about any two-transmission media in detail. [2+5]

   b) What is the role of the Data Link Layer? Explain the Stop-and-Wait ARQ protocol with a suitable diagram. [3+5]

3. a) Discuss about the significance of routing in computer networking? Differentiate between link state routing and distance vector routing. [2+5]

   b) Explain the terms. (4x2)
   i. HTTP and HTTPS ii. DHCP and DNS

4. a) A 7-bit Hamming Code is received as 1011011. Assume even parity and state whether the received code is correct or wrong. If wrong locate the bit in error. [7]

   b) Define congestion control. Explain the Token Bucket Algorithm with an appropriate diagram. (2+6)

   OR

   Briefly explain the various TCP congestion control policies.

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5. a) Why port number is used in networking? Distinguish between TCP and UDP. How is TCP connection established? Explain. [2+2+4]

OR

Define SMTP, POP and IMAP with their port number.

b) What are the drawbacks in IPV4? Which of these drawbacks do IPV6 solve? Explain [2+5]

6. a) Define network security. Discuss various types of attacks in computer networks. How can firewalls prevent such attacks? [1+2+5]

b) Explain the working principle of the RSA algorithm with an example. [7]

7. Write short notes on (any two) (2*5)
   a) Recent Trends in Telecom Technologies: 2G/3G/4G/5G.
   b) Switch
   c) NGN
   d) NAT

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GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Level: Bachelor | Semester – Spring | Year : 2026
Programme: BE: CE VI | Full Marks: 100
Pass Marks: 45
Course: Computer Networks | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Network. Explain the merits and demerits of using it in your own writing. 1+6

   b) Define protocols, interfaces and services. Discuss TCP/IP Layer in brief. 3+5

2. a) Differentiate between Circuit Switching and Packet Switching. Explain the advantages of optical fibre in communication. 2+5

   b) What do you understand by flow control? How does Selective Repeat ARQ overcome the limitations of Go Back N and Stop and wait ARQ. 2+6

OR

What do you understand by error? Explain the working of Hamming Code with example.

3. a) Prepare CRC code for the bit sequences 10011101 given the predetermined divisor x^4+x^3+1. Suppose, while transmitting the fourth bit from the left is inverted. Using CRC explain how can you detect and check error. 8

   b) Why is the world moving towards IPV6 addressing? Explain the working of OSPF. 2+5

OR

Define Public and Private address. Explain the header format of IPV4.

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4. a) PU has four departments Software, Computer, Electronics and Bioinformatics which needs 80,40,20 and 10 computers. Design an IP plan Table for the given subnets using appropriate IP addresses. 8

b) What is a Socket? Differentiate between TCP and UDP. 1+6

5. a) How can we control congestion in the network? Explain about Leaky Bucket algorithm in brief. 2+6

b) Which ports do HTTP and HTTPs use to browse the internet? Explain the working of DHCP. 2+5

6. a) In RSA cryptosystem, A particular A uses two prime numbers 3 and 11 and public key e=7. Compute the private key d. 8

b) Define VPN. Explain different functions of network management. 1+6

7. Write short notes on **(Any Two)**: 5*2

a) Diffie Hellman Key Exchange Protocol
b) NGN
c) ISDN

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United Technical College

2026 Spring

Course Title: Computer Networks **Course Code: CMP 344**

Full Marks: 100 | Pass Marks: 45 Time: 3 Hours

Attempt ALL questions.

Question 1

(a) Define computer networking. Explain how peer-to-peer architecture differs from the traditional client-server architecture in terms of scalability, resource management, and security. [7]

(b) Explain the fundamental topological structures of computer networks, specifically comparing Mesh topology and Hybrid topology. Discuss how recent trends in telecom technologies (such as 4G and 5G) have transformed mobile network backbones compared to legacy 2G systems. [8]

Question 2

(a) Compare the OSI reference model with the TCP/IP protocol suite. Detail the primary functions of each layer in the OSI model [7]

(b) Explain the framing mechanisms and flow control protocols implemented at the Data Link Layer. Compare the efficiency and operational mechanics of Go-Back-N ARQ and Selective-Repeat ARQ. [8]

Question 3

(a) A data stream with bit sequence 1101011011 needs transmission using CRC with generator polynomial $G(x) = x^4 + x + 1$ (10011). Calculate the transmitted frame. Also, explain the operation of CSMA/CD in Ethernet networks. [7]

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(b) Differentiate among Circuit Switching, Packet Switching, and Message Switching mechanisms. A network link has a bandwidth of 10 Mbps and a propagation delay of 20 ms. Calculate the Bandwidth-Delay Product (BDP). [8]

Question 4

(a) An organization receives the IPv4 block 192.168.10.0/24 and needs to form four subnets for different departments: Sales (60 hosts), Engineering (28 hosts), HR (12 hosts), and Point-to-Point WAN link (2 hosts). Design a Variable Length Subnet Masking (VLSM) allocation scheme. For each subnet,

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identify: (i) Subnet Address, (ii) Subnet Mask (in CIDR notation), (iii) Usable IP Range, and (iv) Broadcast Address. [7]

(b) Describe process-to-process delivery at the transport layer using port addressing, multiplexing, and demultiplexing. Compare the TCP segment structure with the UDP datagram structure. [8]

Question 5

(a) Distinguish between Open-Loop and Closed-Loop congestion control mechanisms. Explain the TCP Congestion Control state machine involving Slow Start, Congestion Avoidance, and Fast Recovery phases. [7]

(b) Explain the hierarchical structure of the Domain Name System (DNS). Describe the process of iterative and recursive DNS queries when resolving a domain name into an IP address. [8]

Question 6

(a) Explain the core principles of cryptography, differentiating between Symmetric Key Cryptography (DES) and Asymmetric Key Cryptography (RSA). [7]

(b) Using RSA algorithm, let prime numbers p = 3 and q = 11. Choose public key exponent e = 7. (i) Calculate modulus n and Euler's totient φ(n). (ii) Find the private key exponent d. (iii) Encrypt the plain message M = 2. [8]

Question 7

Short Notes: (Any Two)

a) NGN

b) MPLS

c) VPN

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